

Quantum Amplitude Estimation - Lean

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1 Introduction

2 RotationGates

Herein we find definitions of rotation quantum gates, which have not been introduced in previous repositories.

Definition [2.1]

(Rx)

$$R_x(\theta) = \begin{pmatrix} \cos \frac{\theta}{2} & -i \sin \frac{\theta}{2} \\ -i \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix}$$

Definition [2.2]

(Rz)

$$R_z(\theta) = \begin{pmatrix} e^{-i\frac{\theta}{2}} & 0 \\ 0 & e^{i\frac{\theta}{2}} \end{pmatrix}$$

Definition [2.3]

(Ry)

$$R_y(\theta) = \begin{pmatrix} \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} \\ \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix}$$

3 BHMTK1Tangent

Herein we find a proof from mathematical analysis that will be used later to prove an important theorem regarding Quantum Amplitude Estimation.

Lemma [3.1]

(bhmtTanOddTail_den_pos)

Let $u \in [0, 1)$ and $n \in \mathbb{N}$. Then,

$$\frac{1}{(2(n+2)-1)^2 - u^2} \geq 0.$$

Lemma [3.2]

(bhmtTanOddTail_term_bound)

Let $u \in [0, 1)$ and $n \in \mathbb{N}$. Then,

$$\frac{1}{(2(n+2)-1)^2 - u^2} \leq \frac{1}{2(1+u^2)(n+2)(n+1)}$$

Lemma [3.3]

(bhmtInvNatSuccMulSuccSucc_sum_telescope)

Let $N \in \mathbb{N}$. Then,

$$\sum_{n=0}^{N-1} \frac{1}{(n+1)(n+2)} = 1 - \frac{1}{N+1}.$$

Lemma [3.4]

(bhmtTanOddTail_summable)

Let $u \in [0, 1)$. The following series converges.

$$\sum_{n=0}^{\infty} \frac{1}{(2(n+2)-1)^2 - u^2}.$$

Lemma [3.5]

(bhmtTanOddTail_tsum_le)

Let $u \in [0, 1)$. Then, the next inequality holds true if the series on the left-hand side converges.

$$\sum_{n=0}^{\infty} \frac{1}{(2(n+2)-1)^2 - u^2} \leq \frac{1}{2(1+u^2)}.$$

Lemma [3.6]

(bhmtRealCotTerm_summable)

Let $z \in \mathbb{R} \setminus \mathbb{Z}$. The following series converges.

$$\sum_{n=0}^{\infty} \left(\frac{1}{z - (n+1)} + \frac{1}{z + (n+1)} \right).$$

Lemma [3.7]

(bhmtHalfOneMinus_mem_integerComplement)

Let $u \in [0, 1)$. Then, $\frac{1-u}{2} \notin \mathbb{Z}$.

Lemma [3.8]

(bhmtRealCotTerm_summable_of_bounds)

Let $u \in [0, 1)$. The following series converges.

$$\sum_{n=0}^{\infty} \left(\frac{1}{\frac{1-u}{2} - (n+1)} + \frac{1}{\frac{1-u}{2} + (n+1)} \right).$$

Lemma [3.9]

(bhmtTanGroupedTerm_eq_tail)

Let $u \in [0, 1)$ and $n \in \mathbb{N}$. Then,

$$\frac{1}{\frac{1-u}{2} + (n+1)} + \frac{1}{\frac{1-u}{2} - (n+2)} = \frac{4u}{(2(n+2)-1)^2 - u^2}.$$

Lemma [3.10]

(bhmtTanGroupedTerm_summable)

Let $u \in [0, 1)$. The following series converges.

$$\sum_{n=0}^{\infty} \left(\frac{1}{\frac{1-u}{2} + (n+1)} + \frac{1}{\frac{1-u}{2} - (n+2)} \right).$$

Lemma [3.11]

(bhmtTanGroupedTerm_tsum)

Let $u \in [0, 1)$. The following holds if both series converge.

$$\sum_{n=0}^{\infty} \left(\frac{1}{\frac{1-u}{2} + (n+1)} + \frac{1}{\frac{1-u}{2} - (n+2)} \right) = 4u \sum_{n=0}^{\infty} \frac{1}{(2(n+2)-1)^2 - u^2}.$$

Lemma [3.12]

(bhmtRealCotTerm_finite_regroup)

Let $u \in \mathbb{R}$ and $N \in \mathbb{N}$. Then,

$$\begin{aligned} \sum_{n=0}^N \left(\frac{1}{\frac{1-u}{2} - (n+1)} + \frac{1}{\frac{1-u}{2} + (n+1)} \right) &= \frac{1}{\frac{1-u}{2} - 1} \\ &+ \sum_{n=0}^{N-1} \left(\frac{1}{\frac{1-u}{2} + (n+1)} + \frac{1}{\frac{1-u}{2} - (n+2)} \right) \\ &+ \frac{1}{\frac{1-u}{2} + (N+1)}. \end{aligned}$$

Lemma [3.13]

(bhmtRealCotTerm_endpoint_tendsto_zero)

Let $u \in \mathbb{R}$. Then,

$$\lim_{N \rightarrow \infty} \frac{1}{\frac{1-u}{2} + (N+1)} = 0$$

Lemma [3.14]

(bhmtRealCotTerm_tsum_regroup)

Let $u \in \mathbb{R}$. Assume that the following two series converge.

$$\begin{aligned} \sum_{n=0}^{\infty} \left(\frac{1}{\frac{1-u}{2} - (n+1)} + \frac{1}{\frac{1-u}{2} + (n+1)} \right), \\ \sum_{n=0}^{\infty} \left(\frac{1}{\frac{1-u}{2} + (n+1)} + \frac{1}{\frac{1-u}{2} - (n+2)} \right). \end{aligned}$$

Then,

$$\begin{aligned} \sum_{n=0}^{\infty} \left(\frac{1}{\frac{1-u}{2} - (n+1)} + \frac{1}{\frac{1-u}{2} + (n+1)} \right) &= \frac{1}{\frac{1-u}{2} - 1} \\ &+ \sum_{n=0}^{\infty} \left(\frac{1}{\frac{1-u}{2} + (n+1)} + \frac{1}{\frac{1-u}{2} - (n+2)} \right). \end{aligned}$$

Lemma [3.15]
(bhmtTan_base_poles)
Let $u \in [0, 1)$. Then,

$$\frac{1}{\frac{1-u}{2}} + \frac{1}{\frac{1-u}{2} - 1} = \frac{4u}{1-u^2}.$$

Theorem [3.1]
(bhmtTanPartialFractionIdentity_of_bounds)
Let $u \in [0, 1)$. If the series of the right-hand side converge, the following holds.

$$\pi \tan\left(u \frac{\pi}{2}\right) = \frac{4u}{1-u^2} + \sum_{n=0}^{\infty} \frac{4u}{(2(n+2)-1)^2 - u^2}.$$

Lemma [3.16]
(deriv_bhmtK1TwoNearestCore_eq_neg_log_factor)
Let $x \in (0, \frac{1}{2})$. Then,

$$\begin{aligned} \frac{d}{dx} \left(\frac{\sin^2(\pi x)}{x^2} + \frac{\sin^2(\pi x)}{(1-x)^2} \right) &= -2 \frac{8(1+(1-2x)^2) \cos^2(\pi \frac{1-2x}{2})}{(1-(1-2x)^2)^2} \\ &\quad \times \left(\frac{2(1-2x)}{1+(1-2x)^2} + \frac{4(1-2x)}{1-(1-2x)^2} - \pi \tan\left((1-2x)\frac{\pi}{2}\right) \right). \end{aligned}$$

Lemma [3.17]
(bhmt_log_derivative_expr_nonneg_of_tan_series)
Let $u \in [0, 1)$. Then,

$$\frac{2u}{1+u^2} + \frac{4u}{1-u^2} - \pi \tan\left(u \frac{\pi}{2}\right) \geq 0.$$

Theorem [3.2]
(bhmtK1TwoNearestCore_deriv_nonpos)
For all $x \in (0, \frac{1}{2})$,

$$\frac{d}{dx} \left(\frac{\sin^2(\pi x)}{x^2} + \frac{\sin^2(\pi x)}{(1-x)^2} \right) \leq 0.$$

Theorem [3.3]**(qpeApproxGeometricProbability_two_nearest_lower_bound_k1_of_core_antitone_left)***Let $N \in \mathbb{N}$, $x \in (0, 1)$, $N > 0$. Then,*

$$\frac{\sin^2(\pi x)}{N^2 \sin^2\left(\pi \frac{x}{N}\right)} + \frac{\sin^2(\pi(x-1))}{N^2 \sin^2\left(\pi \frac{x-1}{N}\right)} \geq \frac{8}{\pi^2}.$$

Theorem [3.4]**(qpeCircularPhaseWindowProbability_lower_bound_k1_of_two_adjacent_geometric)***Let $m \in \mathbb{N}$, $\theta, x \in \mathbb{R}$, and $y_0, y_1 \in O_{m,1,\theta}$. Assume $x \in (0, 1)$, and $y_0 \neq y_1$.**Assume also that the following two propositions hold.*

$$\begin{aligned} \|a(m, \theta, y_0)\|^2 &= \frac{\sin^2(2^m \pi \frac{x}{2^m})}{2^{2m} \sin^2(\pi \frac{x}{2^m})}, \\ \|a(m, \theta, y_1)\|^2 &= \frac{\sin^2(2^m \pi \frac{x-1}{2^m})}{2^{2m} \sin^2(\pi \frac{x-1}{2^m})}. \end{aligned}$$

Then,

$$\Pr\{y \in O_{m,1,\theta} | Y\} \geq \frac{8}{\pi^2}.$$

Theorem [3.5]**(qpeCircularPhaseWindowProbability_lower_bound_k1)***Let $m \in \mathbb{N}$ and $\theta \in [0, 1]$. Then,*

$$\Pr\{y \in O_{m,1,\theta} | Y\} \geq \frac{8}{\pi^2}.$$

Theorem [3.6]**(bhmt11CircularWindowBound_proved)***Proposition BHMT11CircularWindowBound in Quantum Phase Estimation section holds.*

4 QuantumPhaseEstimation

Definition [4.1]
(M)

$$M(m) = 2^m.$$

Lemma [4.1]
(m_neZero)
Let $m \in \mathbb{N}$. Then,

$$M(m) \neq 0.$$

Definition [4.2]
(zeroIndex)

$$0(m) = 0.$$

Definition [4.3]
(uniformState)

$$|u(m)\rangle = \frac{1}{\sqrt{2^m}} \sum_{k=0}^{2^m-1} |k\rangle.$$

Definition [4.4]
(phaseState)

$$|p(m, \theta)\rangle = \frac{1}{\sqrt{2^m}} \sum_{k=0}^{2^m-1} e^{2\pi i \theta k} |k\rangle.$$

Definition [4.5]
(qftMatrix)

$$Q(m) = \frac{1}{\sqrt{2^m}} \sum_{k=0}^{2^m-1} \sum_{l=0}^{2^m-1} e^{2\pi i k l / 2^m} |k\rangle \langle l|.$$

Definition [4.6]
(inverseQFTMatrix)

$$IQ(m) = \frac{1}{\sqrt{2^m}} \sum_{k=0}^{2^m-1} \sum_{l=0}^{2^m-1} e^{-2\pi i k l / 2^m} |k\rangle \langle l|.$$

Definition [4.7]
(qpeApproxAmplitude)

$$a(m, \theta, y) = \langle y | IQ(m) | p(m, \theta) \rangle.$$

Definition [4.8]
(qpeApproxOutcomeProbability)

$$Pr\{y \text{ is measured from } IQ(m) | p(m, \theta)\} = ||a(m, \theta, y)||^2.$$

Definition [4.9]
(qpePhaseWindow)

$$\left| \theta - \frac{y}{2^m} \right| \leq \frac{k}{2^m}.$$

Theorem [4.1]
(qpeApproxOutcomeProbability_nonneg)

$$Pr\{y \text{ is measured from } IQ(m) | p(m, \theta)\} \geq 0.$$

Definition [4.10]
(unitPhaseDistance)

$$d(\theta, \rho) = \min\{|\theta - \rho|, |\theta - \rho + 1|, |\theta - \rho - 1|\}.$$

Theorem [4.2]
(unitPhaseDistance_nonneg)

$$d(\theta, \rho) \geq 0.$$

Theorem [4.3]
(unitPhaseDistance_le_abs_sub)

$$d(\theta, \rho) \leq |\theta - \rho|.$$

Theorem [4.4]
(unitPhaseDistance_le_sub_one)

$$d(\theta, \rho) \leq |\theta - \rho - 1|.$$

Theorem [4.5]
(unitPhaseDistance_le_add_one)

$$d(\theta, \rho) \leq |\theta - \rho + 1|.$$

Theorem [4.6]
(unitPhaseDistance_pos_of_mem_Ico_ne)
Let $\theta, \rho \in [0, 1)$ and $\theta \neq \rho$. Then,

$$d(\theta, \rho) > 0.$$

Theorem [4.7]
(unitPhaseDistance_le_half_of_mem_Ico)
Let $\theta, \rho \in [0, 1)$. Then,

$$d(\theta, \rho) \leq \frac{1}{2}.$$

Theorem [4.8]
(unitPhaseDistance_cases)

If $d(\theta, \rho) \leq \epsilon$, then at least one of the following statements is true:

1. $|\theta - \rho| \leq \epsilon,$
2. $|\theta - \rho - 1| \leq \epsilon,$
3. $|\theta - \rho + 1| \leq \epsilon.$

Definition [4.11]
(qpeCircularPhaseWindow)

$$d\left(\theta, \frac{y}{2^m}\right) \leq \frac{k}{2^m}.$$

Definition [4.12]
(qpeCircularPhaseWindowOutcomes)

$$O_{m,k,\theta} = \left\{ y \in \mathbb{Z}_{2^m} \mid d\left(\theta, \frac{y}{2^m}\right) \leq \frac{k}{2^m} \right\}.$$

Definition [4.13]
(qpeCircularPhaseWindowProbability)
Let Y be the event of measuring y from $IQ(m) |p(m, \theta)\rangle$.

$$Pr\{y \in O_{m,k,\theta} | Y\} = \sum_{y \in O_{m,k,\theta}} ||a(m, \theta, y)||^2.$$

Definition [4.14]
(qpeCircularPhaseWindowFailureOutcomes)

$$F_{m,k,\theta} = O_{m,k,\theta}^c.$$

Definition [4.15]
(qpeCircularPhaseWindowFailureProbability)
Let Y be the event of measuring y from $IQ(m) |p(m, \theta)\rangle$.

$$Pr\{y \in F_{m,k,\theta} | Y\} = \sum_{y \in F_{m,k,\theta}} ||a(m, \theta, y)||^2.$$

Theorem [4.9]
(mem_qpeCircularPhaseWindowOutcomes_iff)
Let $m, k \in \mathbb{N}$, $\theta \in \mathbb{R}$, $y \in \mathbb{Z}_{2^m}$. Then, $y \in O_{m,k,\theta}$ if and only if $d\left(\theta, \frac{y}{2^m}\right) \leq \frac{k}{2^m}$.

Theorem [4.10]
(mem_qpeCircularPhaseWindowFailureOutcomes_iff)
Let $m, k \in \mathbb{N}$, $\theta \in \mathbb{R}$, $y \in \mathbb{Z}_{2^m}$. Then, $y \in F_{m,k,\theta}$ if and only if $\neg\left(d\left(\theta, \frac{y}{2^m}\right) \leq \frac{k}{2^m}\right)$.

Definition [4.16]
(floorGridIndexNat)

$$\lfloor 2^m \theta \rfloor.$$

Definition [4.17]
(qpeLowerAdjacentOutcome)

$$\lfloor 2^m \theta \rfloor \pmod{2^m}.$$

Definition [4.18]
(qpeUpperAdjacentOutcome)

$$\lfloor 2^m \theta \rfloor + 1 \pmod{2^m}.$$

Definition [4.19]
(qpeFractionalOffset)

$$2^m \theta - \lfloor 2^m \theta \rfloor.$$

Theorem [4.11]
(floorGridIndexNat_lt_M_of_mem_Ico)
Let $m \in \mathbb{N}$, $\theta \in [0, 1)$. Then,

$$\lfloor 2^m \theta \rfloor < 2^m.$$

Theorem [4.12]
(theta_eq_floorGrid_div_of_qpeFractionalOffset_eq_zero)
Let $m \in \mathbb{N}$, $\theta \geq 0$ and $2^m \theta - \lfloor 2^m \theta \rfloor = 0$. Then,

$$\theta = \frac{\lfloor 2^m \theta \rfloor}{2^m}.$$

Theorem [4.13]
(qpeLowerAdjacentOutcome_mem_circularWindow_one_of_mem_Ico)
Let $m \in \mathbb{N}$, $\theta \in [0, 1)$. Then,

$$\lfloor 2^m \theta \rfloor \in O_{m,1,\theta}.$$

Theorem [4.14]

(qpeUpperAdjacentOutcome_mem_circularWindow_one_of_mem_Ico)

Let $m \in \mathbb{N}$, $\theta \in [0, 1)$. Then,

$$\lfloor 2^m \theta \rfloor + 1 \pmod{2^m} \in O_{m,1,\theta}.$$

Theorem [4.15]

(qpeAdjacentOutcomes_ne_of_pos_m)

Let $m \in \mathbb{N}$, $\theta \in [0, 1)$, and $m > 0$. Then,

$$\lfloor 2^m \theta \rfloor \neq \lfloor 2^m \theta \rfloor + 1 \pmod{2^m}.$$

Theorem [4.16]

(qpeCircularFailure_unitPhaseDistance_gt)

Let $m, k \in \mathbb{N}$, $\theta \in \mathbb{R}$, and $y \in F_{m,k,\theta}$. Then,

$$d\left(\theta, \frac{y}{2^m}\right) > \frac{k}{2^m}.$$

Theorem [4.17]

(qpeCircularFailure_phase_error_representative)

Let $m \in \mathbb{N}$, $\theta \in [0, 1]$, $k \in \mathbb{N}$, and $y \in F_{m,k,\theta}$. There exists δ such that at least one of the following statements is true:

1. $\delta = \theta - \frac{y}{2^m},$
2. $\delta = \theta - \frac{y}{2^m} - 1,$
3. $\delta = \theta - \frac{y}{2^m} + 1.$

Moreover,

$$\frac{k}{2^m} < |\delta| \leq \frac{1}{2}.$$

Theorem [4.18]

(fin_eq_left_or_right_candidate_of_abs_sub_mem_Ico)

Let $N \in \mathbb{N}$ and $N > 0$. Let $x \in \mathbb{R}$, $i \in \mathbb{N}$, $y \in \mathbb{Z}_N$, and $z \in \mathbb{Z}$. Assume:

1. $y = z \pmod{N}$,
2. $i \leq |x - z| < i + 1$.

Then, at least one of the following holds:

$$\begin{aligned} y &= \lfloor x \rfloor - i \pmod{N}, \\ y &= \lceil x \rceil + i \pmod{N}. \end{aligned}$$

Definition [4.20]
(qpeCircularDistanceBucket)

$$\left\lfloor 2^m d\left(\theta, \frac{y}{2^m}\right) \right\rfloor.$$

Theorem [4.19]
(qpeCircularDistanceBucket_mem_tail_of_failure)

Let $m, k \in \mathbb{N}$, and $\theta \in [0, 1]$. Let $k > 0$ and $y \in F_{m,k,\theta}$. Then,

$$k - 1 < \left\lfloor 2^m d\left(\theta, \frac{y}{2^m}\right) \right\rfloor \leq 2^m.$$

Theorem [4.20]
(qpeCircularFailure_phase_error_representative_with_unitDistance)

Let $m \in \mathbb{N}$, $\theta \in [0, 1]$, $k \in \mathbb{N}$, and $y \in F_{m,k,\theta}$. There exists δ such that at least one of the following statements is true:

1. $\delta = \theta - \frac{y}{2^m}$,
2. $\delta = \theta - \frac{y}{2^m} - 1$,
3. $\delta = \theta - \frac{y}{2^m} + 1$.

Moreover,

$$\frac{k}{2^m} < |\delta| \leq \frac{1}{2},$$

and

$$d\left(\theta, \frac{y}{2^m}\right) \leq |\delta|.$$

Definition [4.21]

(bhmt11SuccessProbability)

If $k = 1$, then

$$\frac{8}{\pi^2}.$$

If $k \neq 1$, then,

$$1 - \frac{1}{2(k-1)}.$$

Definition [4.22]

(BHMT11CircularWindowBound)

Let $m, k \in \mathbb{N}$, $\theta \in [0, 1]$. Assume also that $k > 0$.

If $k = 1$,

$$\frac{8}{\pi^2} \leq \Pr\{y \in O_{m,k,\theta} | Y\}.$$

If $k > 1$,

$$1 - \frac{1}{2(k-1)} \leq \Pr\{y \in O_{m,k,\theta} | Y\}.$$

Theorem [4.21]

(phaseState_add_one)

Let $m \in \mathbb{N}$ and $\theta \in \mathbb{R}$. Then,

$$|p(m, \theta + 1)\rangle = |p(m, \theta)\rangle.$$

Theorem [4.22]

(qpeApproxAmplitude_add_one)

Let $m \in \mathbb{N}$, $\theta \in \mathbb{R}$, and $y \in \mathbb{Z}_{2^m}$. Then,

$$a(m, \theta + 1, y) = a(m, \theta, y).$$

Theorem [4.23]

(qpeApproxOutcomeProbability_add_one)

Let $m \in \mathbb{N}$, $\theta \in \mathbb{R}$, and $y \in \mathbb{Z}_{2^m}$. Then,

$$||a(m, \theta + 1, y)||^2 = ||a(m, \theta, y)||^2.$$

Theorem [4.24]**(qpeApproxOutcomeProbability_one_sub)***Let $m \in \mathbb{N}$, $\theta \in \mathbb{R}$, and $y \in \mathbb{Z}_{2^m}$. Then,*

$$||a(m, 1 - \theta, y)||^2 = ||a(m, -\theta, y)||^2.$$

Definition [4.23]**(qpeApproxGeometricProbability)***If $\delta = 0$, then*

$$1.$$

If $\delta \neq 0$, then

$$\frac{\sin^2(N\pi\delta)}{N^2 \sin^2(\pi\delta)}.$$

Theorem [4.25]**(qpeCircularFailureProbability_le_of_bucket_tail)***Let $m, k, n \in \mathbb{N}$ and $k > 1$. Let $f : \mathbb{Z}_{2^m} \rightarrow \mathbb{N}$ such that the following propositions hold:**1. For all $y \in F_{m,k,\theta}$,*

$$k - 1 < f(y) \leq n.$$

2. For all $y \in F_{m,k,\theta}$,

$$||a(m, \theta, y)||^2 \leq \frac{1}{4f(y)^2}.$$

3. For all $i \in (k - 1, n] \cap \mathbb{N}$,

$$|F_{m,k,\theta} \cap \{y \in \mathbb{Z}_{2^m} | f(y) = i\}| \leq 2.$$

Then,

$$Pr\{y \in F_{m,k,\theta} | Y\} \leq \frac{1}{2(k - 1)}.$$

Definition [4.24]

(bhmtK1TwoNearestCore)

$$\frac{\sin^2(\pi x)}{x^2} + \frac{\sin^2(\pi x)}{(1-x)^2}.$$

Theorem [4.26]

(bhmtK1TwoNearestCore_symm)

bhmtK1TwoNearestCore is symmetric over $\frac{1}{2}$.

Theorem [4.27]

(bhmtK1TwoNearestCore_half)

bhmtK1TwoNearestCore is 8 when $x = \frac{1}{2}$.

Theorem [4.28]

(bhmtK1TwoNearestCore_ge_eight_of_antitone_left)

If bhmtK1TwoNearestCore is non-increasing on $(0, \frac{1}{2})$ and $x \in (0, 1)$, then

$$\frac{\sin^2(\pi x)}{x^2} + \frac{\sin^2(\pi x)}{(1-x)^2} \geq 8.$$

Theorem [4.29]

(deriv_bhmtK1TwoNearestCore)

Let $x \in \mathbb{R}$ and $x \neq 0, 1$. Then,

$$\frac{d}{dx} \left(\frac{\sin^2(\pi x)}{x^2} + \frac{\sin^2(\pi x)}{(1-x)^2} \right) (x) = \frac{\pi \sin(2\pi x)}{x^2} - \frac{2 \sin^2(\pi x)}{x^3} + \frac{\pi \sin(2\pi x)}{(1-x)^2} + \frac{2 \sin^2(\pi x)}{(1-x)^3}.$$

Theorem [4.30]

(bhmtK1TwoNearestCore_antitone_left_of_deriv_nonpos)

If the derivative of bhmtK1TwoNearestCore is non-positive on $(0, \frac{1}{2})$, then

bhmtK1TwoNearestCore is non-increasing on $(0, \frac{1}{2})$.

Theorem [4.31]

(qpeApproxGeometricProbability_neg)

qpeApproxGeometricProbability is symmetric on δ over 0.

Theorem [4.32]

(qpeApproxGeometricProbability_abs)

qpeApproxGeometricProbability is symmetric on δ over 0.

Theorem [4.33]

(qpeApproxGeometricProbability_add_one_of_sin_ne_zero)

Let $N \in \mathbb{N}$ and $\delta \in \mathbb{R}$. Assume that

$$\sin(\pi\delta) \neq 0.$$

Then,

$$\frac{\sin^2(N\pi(\delta + 1))}{N^2 \sin^2(\pi\delta)} = \frac{\sin^2(N\pi\delta)}{N^2 \sin^2(\pi\delta)}.$$

Theorem [4.34]

(qpeApproxGeometricProbability_sub_one_of_sin_ne_zero)

Let $N \in \mathbb{N}$ and $\delta \in \mathbb{R}$. Assume that

$$\sin(\pi\delta) \neq 0.$$

Then,

$$\frac{\sin^2(N\pi(\delta - 1))}{N^2 \sin^2(\pi\delta)} = \frac{\sin^2(N\pi\delta)}{N^2 \sin^2(\pi\delta)}.$$

Theorem [4.35]

(qpeApproxGeometricProbability_unitPhaseDistance_eq_of_sin_ne_zero)

Let $N \in \mathbb{N}$ and $\theta, \rho \in \mathbb{R}$ such that

$$\sin(\pi(\theta - \rho)) \neq 0.$$

Then,

$$\frac{\sin^2(N\pi d(\theta, \rho))}{N^2 \sin^2(\pi d(\theta, \rho))} = \frac{\sin^2(N\pi(\theta - \rho))}{N^2 \sin^2(\pi(\theta - \rho))}.$$

Theorem [4.36]

(qpeApproxGeometricProbability_lower_bound_scaled)

Let $N \in \mathbb{N}$ and $x \in (0, 1)$, $N > 0$, and $\delta = \frac{x}{N}$. Then,

$$\frac{\sin^2(\pi x)}{\pi^2 x^2} \leq \frac{\sin^2(N\pi\delta)}{N^2 \sin^2(\pi\delta)}.$$

Theorem [4.37]

(qpeApproxGeometricProbability_two_nearest_lower_bound_of_core)

Let $N \in \mathbb{N}$ and $x \in (0, 1)$, and $N > 0$. Then,

$$\frac{1}{\pi^2} \left(\frac{\sin^2(\pi x)}{x^2} + \frac{\sin^2(\pi x)}{(1-x)^2} \right) \leq \frac{\sin^2(\pi x)}{N^2 \sin^2\left(\pi \frac{x}{N}\right)} + \frac{\sin^2(\pi(x-1))}{N^2 \sin^2\left(\pi \frac{x-1}{N}\right)}.$$

Theorem [4.38]

(qpeCircularPhaseWindowProbability_lower_bound_two_outcomes)

Let $m, k \in \mathbb{N}$, $\theta, c \in \mathbb{R}$, $y_0, y_1 \in \mathbb{Z}_{2^m}$. Let also $y_0, y_1 \in O_{m,k,\theta}$ and $y_0 \neq y_1$. Assume the following.

$$c \leq \|a(m, \theta, y_0)\|^2 + \|a(m, \theta, y_1)\|^2.$$

Then,

$$c \leq \Pr\{y \in O_{m,k,\theta} | Y\}.$$

Definition [4.25]

(controlledPowerMatrix)

Let U be a square matrix of dimension n , and $m \in \mathbb{N}$.

$$\Lambda(U, m) = \text{diag} \left(I, U, \dots, U^{2^m-1} \right).$$

Theorem [4.39]

(qftMatrix_mul_basisState)

Let $m \in \mathbb{N}$ and $y \in \mathbb{Z}_{2^m}$. Then,

$$Q(m) |y\rangle = |p\left(m, \frac{y}{2^m}\right)\rangle$$

Theorem [4.40]

(controlledPowerMatrix_mul_uniform_of_real_power_action)

Let $n, m \in \mathbb{N}$. Let U be a square matrix of dimension n , ψ a vector of dimension n , and $\theta \in \mathbb{R}$. Assume that, for all $k \in \mathbb{Z}_{2^m}$, the following holds.

$$U^k \psi = e^{2\pi i k \theta} \psi.$$

Then,

$$\Lambda(U, m)(|u(m)\rangle \otimes \psi) = |p(m, \theta)\rangle \otimes \psi.$$

Theorem [4.41]

(controlledPowerMatrix_mul_uniform_of_real_eigenphase)

Let $n, m \in \mathbb{N}$. Let U be a square matrix of dimension n , ψ a vector of dimension n , and $\theta \in \mathbb{R}$. Assume also that

$$U\psi = e^{2\pi i\theta}\psi.$$

Then,

$$\Lambda(U, m)(|u(m)\rangle \otimes \psi) = |p(m, \theta)\rangle \otimes \psi.$$

Theorem [4.42]

(phaseState_isNormalized)

Let $m \in \mathbb{N}$ and $\theta \in \mathbb{R}$. Then, $|p(m, \theta)\rangle$ is normalized.

Theorem [4.43]

(inverseQFTMatrix_eq_adjoint_qftMatrix)

Let $m \in \mathbb{N}$. Then,

$$IQ(m) = Q^\dagger(m)$$

Theorem [4.44]

(qft_phase_orthogonality)

Let $m \in \mathbb{N}$ and $r, c \in \mathbb{Z}_{2^m}$. Then,

$$\sum_{k=0}^{2^m-1} e^{2\pi ikr/2^m} e^{2\pi ikc/2^m} = \begin{cases} 2^m & r = c \\ 0 & \text{otherwise} \end{cases}.$$

Theorem [4.45]

(qft_normalization_factor)

Let $m \in \mathbb{N}$. Then,

$$\frac{1}{\sqrt{2^m}} \frac{1}{\sqrt{2^m}} = \frac{1}{2^m}.$$

Theorem [4.46]

(qpeApproxAmplitude_eq_normalized_phase_sum)

Let $m \in \mathbb{N}$, $\theta \in \mathbb{R}$, and $y \in \mathbb{Z}_{2^m}$. Then,

$$a(m, \theta, y) = \frac{1}{2^m} \sum_{k=0}^{2^m-1} e^{2\pi i k \left(\theta - \frac{y}{2^m}\right)}.$$

Theorem [4.47]

(phase_fin_sum_normSq)

Let $N \in \mathbb{N}$, $\delta \in \mathbb{R}$. Let also $\sin(\pi\delta) \neq 0$. Then,

$$\left\| \sum_{k=0}^{N-1} e^{2\pi i k \delta} \right\|^2 = \frac{\sin^2(\pi N \delta)}{\sin^2(\pi \delta)}.$$

Theorem [4.48]

(qftMatrix_adjoint_mul_self)

Let $m \in \mathbb{N}$. Then,

$$Q^\dagger(m)Q(m) = I.$$

Theorem [4.49]

(qftMatrix_isUnitary)

Let $m \in \mathbb{N}$. $Q(m)$ is unitary.

Theorem [4.50]

(inverseQFTMatrix_isUnitary)

Let $m \in \mathbb{N}$. $IQ(m)$ is unitary.

Theorem [4.51]

(inverseQFTMatrix_mul_phaseState)

Let $m \in \mathbb{N}$ and $y \in \mathbb{Z}_{2^m}$. Then,

$$IQ(m) \left| p \left(m, \frac{y}{2^m} \right) \right\rangle = |y\rangle.$$

Theorem [4.52]

(qpeApproxOutcomeProbability_total)

Let $m \in \mathbb{N}$ and $\theta \in \mathbb{R}$. Then,

$$\sum_{y=0}^{2^m-1} ||a(m, \theta, y)||^2 = 1.$$

Theorem [4.53]

(qpeApproxOutcomeProbability_eq_geometric_of_sin_ne_zero)

Let $m \in \mathbb{N}$, $\theta \in \mathbb{R}$ and $y \in \mathbb{Z}_{2^m}$. Assume that

$$\sin\left(\pi\left(\theta - \frac{y}{2^m}\right)\right) \neq 0.$$

Then,

$$||a(m, \theta, y)||^2 = \frac{\sin^2(2^m \pi \left(\theta - \frac{y}{2^m}\right))}{2^{2m} \sin^2(\pi \left(\theta - \frac{y}{2^m}\right))}.$$

Theorem [4.54]

(qpeApproxOutcomeProbability_eq_geometric_of_nearest)

Let $m \in \mathbb{N}$, $\theta \in \mathbb{R}$ and $y \in \mathbb{Z}_{2^m}$. Assume the following.

$$\left|\theta - \frac{y}{2^m}\right| \leq \frac{1}{2^{m+1}}.$$

If $\theta - \frac{y}{2^m} \neq 0$,

$$||a(m, \theta, y)||^2 = \frac{\sin^2(2^m \pi \left(\theta - \frac{y}{2^m}\right))}{2^{2m} \sin^2(\pi \left(\theta - \frac{y}{2^m}\right))}.$$

And otherwise,

$$||a(m, \theta, y)||^2 = 1.$$

Theorem [4.55]

(qpeApproxGeometricProbability_upper_bound_sin_den)

Let $N \in \mathbb{N}$ and $\delta \in \mathbb{R}$. Assume that $N > 0$ and that the following holds.

$$\sin(\pi\delta) \neq 0.$$

Then,

$$\frac{\sin^2(N\pi\delta)}{N^2 \sin^2(\pi\delta)} \leq \frac{1}{N^2 \sin^2(\pi\delta)}.$$

Theorem [4.56]

(qpeApproxGeometricProbability_upper_bound_unitDistance)

Let $N \in \mathbb{N}$ and $\delta \in (0, \frac{1}{2}]$. Assume that $N > 0$. Then,

$$\frac{\sin^2(N\pi\delta)}{N^2 \sin^2(\pi\delta)} \leq \frac{1}{4N^2\delta^2}.$$

Theorem [4.57]

(qpeApproxGeometricProbability_upper_bound_of_sin_lower)

Let $N, j \in \mathbb{N}$ and $\delta \in \mathbb{R}$. Let also $N, j > 0$, $\sin(\pi\delta) \neq 0$ and

$$\frac{2j}{N} \leq |\sin(\pi\delta)|.$$

Then,

$$\frac{\sin^2(N\pi\delta)}{N^2 \sin^2(\pi\delta)} \leq \frac{1}{4j^2}.$$

Corollary

(qpeApproxOutcomeProbability_upper_bound_of_sin_lower)

The same theorem holds for $\delta = \theta - \frac{y}{2^m}$, for all $\theta \in \mathbb{R}$ and $y \in \mathbb{Z}_{2^m}$.

Theorem [4.58]

(qpeApproxOutcomeProbability_failure_upper_bound_distanceBucket)

Let $m, k \in \mathbb{N}$, $\theta \in [0, 1]$, and $y \in \mathbb{Z}_{2^m}$. Let $k > 1$ and $y \in F_{m,k,\theta}$. Then,

$$||a(m, \theta, y)||^2 \leq \frac{1}{4 \lfloor 2^m d \left(\theta, \frac{y}{2^m} \right) \rfloor^2}.$$

Theorem [4.59]

(qpeCircularDistanceBucket_eq_imp_left_or_right_candidate)

Let $m, i \in \mathbb{N}$, $\theta \in \mathbb{R}$, and $y \in \mathbb{Z}_{2^m}$. Assume that

$$\lfloor 2^m d\left(\theta, \frac{y}{2^m}\right) \rfloor = i.$$

Then, at least one of the following two statements holds.

1. $y = \lfloor 2^m \theta \rfloor - i \pmod{2^m}$,
2. $y = \lceil 2^m \theta \rceil + i \pmod{2^m}$.

Theorem [4.60]

(qpeCircularDistanceBucket_filter_card_le_two)

Let $m, k, i \in \mathbb{N}$ and $\theta \in \mathbb{R}$. Then,

$$|F_{m,k,\theta} \cap \{y \mid \lfloor 2^m d\left(\theta, \frac{y}{2^m}\right) \rfloor = i\}| \leq 2.$$

Theorem [4.61]

(qpeCircularFailureProbability_le_k_gt_one)

Let $m, k \in \mathbb{N}$ and $\theta \in [0, 1]$. Let $k > 1$. Then,

$$\Pr\{y \in F_{m,k,\theta} \mid Y\} \leq \frac{1}{2(k-1)}.$$

Theorem [4.62]

(qpeCircularPhaseWindowProbability_lower_bound_k_gt_one)

Let $m, k \in \mathbb{N}$ and $\theta \in [0, 1]$. Let $k > 1$. Then,

$$1 - \frac{1}{2(k-1)} \leq \Pr\{y \in O_{m,k,\theta} \mid Y\}.$$

Theorem [4.63]

(qpeApproxOutcomeProbability_lowerAdjacent_eq_one_of_fractionalOffset_eq_zero)

Let $m \in \mathbb{N}$ and $\theta \in [0, 1]$. Let also

$$2^m \theta - \lfloor 2^m \theta \rfloor = 0.$$

Then,

$$||a(m, \theta, 2^m \theta)||^2 = 1$$

Theorem [4.64]

(qpeCircularPhaseWindowProbability_lower_bound_k1_exactGrid_of_mem_Ico)

Let $m \in \mathbb{N}$ and $\theta \in [0, 1)$. Let also

$$2^m \theta - \lfloor 2^m \theta \rfloor = 0.$$

Then,

$$\frac{8}{\pi^2} \leq \Pr\{y \in O_{m,1,\theta} | Y\}.$$

Theorem [4.65]

(qpeApproxOutcomeProbability_lowerAdjacent_eq_geometric_fractionalOffset)

Let $m \in \mathbb{N}$ and $\theta \in [0, 1)$, and $\delta = \frac{2^m \theta - \lfloor 2^m \theta \rfloor}{2^m}$. Assume that

$$2^m \theta - \lfloor 2^m \theta \rfloor > 0.$$

Then,

$$||a(m, \theta, \lfloor 2^m \theta \rfloor)||^2 = \frac{\sin^2(2^m \pi \delta)}{2^{2m} \sin^2(\pi \delta)}.$$

Theorem [4.66]

(qpeApproxOutcomeProbability_upperAdjacent_eq_geometric_fractionalOffset)

Let $m \in \mathbb{N}$ and $\theta \in [0, 1)$, and $\delta = \frac{2^m \theta - \lfloor 2^m \theta \rfloor}{2^m} - \frac{1}{2^m}$. Assume that

$$2^m \theta - \lfloor 2^m \theta \rfloor > 0.$$

If $\delta \neq 0$, then

$$||a(m, \theta, \lfloor 2^m \theta \rfloor + 1 \pmod{2^m})||^2 = \frac{\sin^2(2^m \pi \delta)}{2^{2m} \sin^2(\pi \delta)}.$$

And if $\delta = 0$, then

$$||a(m, \theta, \lfloor 2^m \theta \rfloor + 1 \pmod{2^m})||^2 = 1.$$

5 QuantumLibraryBridge

Definition [5.1]
(qaePlaneVector)

$$\cos(\theta) |0\rangle + \sin(\theta) |1\rangle .$$

6 GroverQPEBridge

Definition [6.1]
(ketMinusI)

$$|-i\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i |1\rangle).$$

Definition [6.2]
(ketPlusI)

$$|+i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i |1\rangle).$$

Theorem [6.1]
(Ry_ketMinusI_eigen)
Let $\theta \in \mathbb{R}$. Then,

$$R_y(4\theta) |-i\rangle = e^{2i\theta} |-i\rangle .$$

Theorem [6.2]
(Ry_ketPlusI_eigen)
Let $\theta \in \mathbb{R}$. Then,

$$R_y(4\theta) |+i\rangle = e^{-2i\theta} |+i\rangle .$$

7 GroverQAESuperposition

Definition [7.1]
(qaeCoeffPlus)

$$\frac{1}{\sqrt{2}}(\cos(\theta) + i \sin(\theta)).$$

Definition [7.2]
(qaeCoeffMinus)

$$\frac{1}{\sqrt{2}}(\cos(\theta) - i \sin(\theta)).$$

Theorem [7.1]
(qaeCoeffPlus_normSq)

$$\left\| \frac{1}{\sqrt{2}}(\cos(\theta) + i \sin(\theta)) \right\|^2 = \frac{1}{2}.$$

Theorem [7.2]
(qaeCoeffMinus_normSq)

$$\left\| \frac{1}{\sqrt{2}}(\cos(\theta) - i \sin(\theta)) \right\|^2 = \frac{1}{2}.$$

Theorem [7.3]
(qaePlaneVector_eq_eigen_superposition)

$$\cos(\theta) |0\rangle + \sin(\theta) |1\rangle = \frac{1}{\sqrt{2}}(\cos(\theta) + i \sin(\theta)) |-i\rangle + \frac{1}{\sqrt{2}}(\cos(\theta) - i \sin(\theta)) |+i\rangle.$$

Theorem [7.4]
(qpeOutputStateConcrete_linear_combination)

Let $n, m \in \mathbb{N}$, U a square matrix of dimension n , ψ, ϕ vectors of dimension n , and $a, b \in \mathbb{C}$. Then,

$$\begin{aligned}
& (IQ(m) \otimes I_n) \times \Lambda(U, m) \times (|u(m)\rangle \otimes (a\psi + b\phi)) \\
& = a(IQ(m) \otimes I_n) \times \Lambda(U, m) \times (|u(m)\rangle \otimes \psi) \\
& + b(IQ(m) \otimes I_n) \times \Lambda(U, m) \times (|u(m)\rangle \otimes \phi).
\end{aligned}$$

Theorem [7.5]

(qpeOutput_groverPlane_initial_eq_eigenphase_superposition)

Let $m \in \mathbb{N}$ and $\theta \in \mathbb{R}$. Then,

$$\begin{aligned}
& (IQ(m) \otimes I_2) \times \Lambda(R_y(4\theta), m) \times (|u(m)\rangle \otimes (\cos(\theta) |0\rangle + \sin(\theta) |1\rangle)) \\
& = \frac{1}{\sqrt{2}}(\cos(\theta) + i \sin(\theta)) IQ(m) \times (|p(m, \frac{\theta}{\pi})\rangle \otimes |-i\rangle) \\
& + \frac{1}{\sqrt{2}}(\cos(\theta) - i \sin(\theta)) IQ(m) \times (|p(m, -\frac{\theta}{\pi})\rangle \otimes |+i\rangle).
\end{aligned}$$

Definition [7.3]

(qpeCountingMarginal)

Let $n, m \in \mathbb{N}$, ψ a vector of dimension $2^m \times n$, and $y \in \mathbb{Z}_{2^m}$.

$$\sum_{j=0}^{n-1} ||\psi_{j+ny}||^2.$$

Theorem [7.6]

(qpeCountingMarginal_groverPlane_initial)

Let $m \in \mathbb{N}$, $\theta \in \mathbb{R}$, and $y \in \mathbb{Z}_{2^m}$. Let also

$$v = (IQ(m) \otimes I_2) \times \Lambda(R_y(4\theta), m) \times (|u(m)\rangle \otimes (\cos(\theta) |0\rangle + \sin(\theta) |1\rangle)).$$

Then,

$$\sum_{j=0}^1 ||v_{j+2y}||^2 = \frac{1}{2} ||a(m, \frac{\theta}{\pi}, y)||^2 + \frac{1}{2} ||a(m, -\frac{\theta}{\pi}, y)||^2.$$

8 QuantumAmplitudeEstimation

Definition [8.1]

(amplitudeFromAngle)

Let $\theta \in \mathbb{R}$.

$$\sin^2(\theta).$$

Definition [8.2]

(estAmpEstimate)

Let $M, y \in \mathbb{N}$.

$$\sin^2\left(\pi \frac{y}{M}\right).$$

Definition [8.3]

(phaseErrorRadius)

Let $M, k \in \mathbb{N}$.

$$\pi \frac{k}{M}.$$

Definition [8.4]

(theorem12ErrorBound)

Let $a \in \mathbb{R}$ and $M, k \in \mathbb{N}$.

$$2\pi \frac{k}{M} \sqrt{a(1-a)} + \pi^2 \frac{k^2}{M^2}.$$

Definition [8.5]

(theorem12SuccessProbability)

Let $k \in \mathbb{N}$. If $k = 1$, then

$$\frac{8}{\pi^2}.$$

If $k \neq 1$, then

$$1 - \frac{1}{2(k-1)}.$$

Theorem [8.1]

(paperLemma7)

Let $a, \theta, \hat{\theta}, \epsilon$. Let also $\epsilon \geq 0$, $a = \sin^2(\theta)$. Assume that

$$|\theta - \hat{\theta}| \leq \epsilon.$$

Then,

$$|\sin^2(\hat{\theta}) - a| \leq 2\epsilon\sqrt{a(1-a)} + \epsilon^2.$$

Theorem [8.2]

(PaperTheorem11)

Let $m, k \in \mathbb{N}$ and $\theta \in [0, 1]$. The following four statements hold:

1. If $2^m\theta - \lfloor 2^m\theta \rfloor = 0$ and $\theta \neq 1$, then

$$||a(m, \theta, 2^m\theta)||^2 = 1.$$

2. If $2^m\theta - \lfloor 2^m\theta \rfloor \neq 0$, let $x \in \mathbb{Z}_{2^m}$, $\theta \neq 1$, and $\delta = d(\theta, \frac{x}{2^m})$. Then,

$$||a(m, \theta, x)||^2 = \frac{\sin^2(2^m\pi\delta)}{2^{2m}\sin^2(\pi\delta)} \leq \frac{1}{2^{2m+2}\delta^2}.$$

3. If $k > 1$, then

$$Pr\{y \in O_{m,k,\theta}|Y\} \geq 1 - \frac{1}{2(k-1)}.$$

4. If $k = 1$ and $2^m > 2$, then

$$Pr\{y \in O_{m,k,\theta}|Y\} \geq \frac{8}{\pi^2}.$$

Definition [8.6]

(phaseOracle)

Let $n \in \mathbb{N}$ and $f : \mathbb{Z}_{2^n} \rightarrow \{0, 1\}$.

$$S_f^n |k\rangle = \begin{cases} -|k\rangle, & f(k) = 1 \\ |k\rangle, & f(k) = 0 \end{cases}.$$

Definition [8.7]

(zeroReflection)

Let $n \in \mathbb{N}$.

$$S_0^n |k\rangle = \begin{cases} -|k\rangle, & k = 0 \\ |k\rangle, & \text{otherwise} \end{cases}.$$

Theorem [8.3]

(zeroReflection_mul_zero_basis)

Let $n \in \mathbb{N}$. Then,

$$S_0^n |0\rangle = -|0\rangle. \tag{1}$$

Definition [8.8]

(paperGroverOperator)

Let $n \in \mathbb{N}$, A a square matrix of dimension 2^n , and $f : \mathbb{Z}_{2^n} \rightarrow \{0, 1\}$.

$$G_f(A) = -AS_0^n A^\dagger S_f^n.$$

Theorem [8.4]

(zeroReflection_mul_vector)

Let $n \in \mathbb{N}$ and v be a vector of dimension 2^n . Then,

$$S_0^n v = v - 2v_0 |0\rangle.$$

Theorem [8.5]

(conjugatedZeroReflection_mul_vector)

Let $n \in \mathbb{N}$, A a unitary matrix of dimension 2^n , and v a vector of dimension 2^n . Then,

$$AS_0^n A^\dagger v = v - 2(A^\dagger v)_0 A|0\rangle.$$

Theorem [8.6]

(phaseOracle_mul_of_bad_support)

Let $n \in \mathbb{N}$, $f : \mathbb{Z}_{2^n} \rightarrow \{0, 1\}$, and v a vector of dimension 2^n . Assume that $v_k = 0$ holds for all k such that $f(k) = 1$. Then,

$$S_f^n v = v.$$

Theorem [8.7]

(phaseOracle_mul_of_good_support)

Let $n \in \mathbb{N}$, $f : \mathbb{Z}_{2^n} \rightarrow \{0, 1\}$, and v a vector of dimension 2^n . Assume that $v_k = 0$ holds for all k such that $f(k) = 0$. Then,

$$S_f^n v = -v.$$

Theorem [8.8]

(bad_good_support_orthogonal)

Let $n \in \mathbb{N}$, $f : \mathbb{Z}_{2^n} \rightarrow \{0, 1\}$, and ψ_0, ψ_1 be vector of dimension 2^n . Assume that $\psi_{0k} = 0$ for all k such that $f(k) = 1$, while $\psi_{1k} = 0$ for all k such that $f(k) = 0$. Then,

$$\langle \psi_0 | \psi_1 \rangle = 0.$$

Corollary

(good_bad_support_orthogonal)

Also,

$$\langle \psi_1 | \psi_0 \rangle = 0.$$

Theorem [8.9]

(badComponent_overlap_of_decomposition)

Let $n \in \mathbb{N}$, A be a square matrix of dimension 2^n , ψ_0, ψ_1 vectors of dimension 2^n , and $\theta \in \mathbb{R}$. Assume that

$$A|0\rangle = \cos(\theta)\psi_0 + \sin(\theta)\psi_1.$$

Assume also that ψ_0 is normalized and that $\langle\psi_1|\psi_0\rangle = 0$. Then,

$$(A^\dagger\psi_0)_0 = \cos\theta.$$

Theorem [8.10]

(goodComponent_overlap_of_decomposition)

Let $n \in \mathbb{N}$, A be a square matrix of dimension 2^n , ψ_0, ψ_1 vectors of dimension 2^n , and $\theta \in \mathbb{R}$. Assume that

$$A|0\rangle = \cos(\theta)\psi_0 + \sin(\theta)\psi_1.$$

Assume also that ψ_1 is normalized and that $\langle\psi_0|\psi_1\rangle = 0$. Then,

$$(A^\dagger\psi_1)_0 = \sin\theta.$$

Theorem [8.11]

(paperGroverOperator_mul_badComponent)

Let $n \in \mathbb{N}$, A be a unitary matrix of dimension 2^n , ψ_0, ψ_1 vectors of dimension 2^n , $f : \mathbb{Z}_{2^n} \rightarrow \{0, 1\}$, and $\theta \in \mathbb{R}$. Assume that $\psi_{0_k} = 0$ for all k such that $f(k) = 1$, and that

$$A|0\rangle = \cos(\theta)\psi_0 + \sin(\theta)\psi_1.$$

Assume also that

$$(A^\dagger\psi_0)_0 = \cos(\theta).$$

Then,

$$G_f(A)\psi_0 = \cos(2\theta)\psi_0 + \sin(2\theta)\psi_1.$$

Theorem [8.12]**(paperGroverOperator_mul_goodComponent)**

Let $n \in \mathbb{N}$, A be a unitary matrix of dimension 2^n , ψ_0, ψ_1 vectors of dimension 2^n , $f : \mathbb{Z}_{2^n} \rightarrow \{0, 1\}$, and $\theta \in \mathbb{R}$. Assume that $\psi_{1k} = 0$ for all k such that $f(k) = 0$, and that

$$A|0\rangle = \cos(\theta)\psi_0 + \sin(\theta)\psi_1.$$

Assume also that

$$(A^\dagger\psi_1)_0 = \sin(\theta).$$

Then,

$$G_f(A)\psi_1 = -\sin(2\theta)\psi_0 + \cos(2\theta)\psi_1.$$

Theorem [8.13]**(groverRotation_eigenMinus)**

Let $n \in \mathbb{N}$, A be a square matrix of dimension 2^n , ψ_0, ψ_1 be vectors of dimension 2^n , and $\theta \in \mathbb{R}$. Assume that the following two propositions hold.

$$\begin{aligned} A\psi_0 &= \cos(2\theta)\psi_0 + \sin(2\theta)\psi_1, \\ A\psi_1 &= -\sin(2\theta)\psi_0 + \cos(2\theta)\psi_1. \end{aligned}$$

Then,

$$\frac{1}{\sqrt{2}}A(\psi_0 - i\psi_1) = \frac{1}{\sqrt{2}}e^{2\theta i}(\psi_0 - i\psi_1).$$

Theorem [8.14]**(groverRotation_eigenPlus)**

Let $n \in \mathbb{N}$, A be a square matrix of dimension 2^n , ψ_0, ψ_1 be vectors of dimension 2^n , and $\theta \in \mathbb{R}$. Assume that the following two propositions hold.

$$\begin{aligned} A\psi_0 &= \cos(2\theta)\psi_0 + \sin(2\theta)\psi_1, \\ A\psi_1 &= -\sin(2\theta)\psi_0 + \cos(2\theta)\psi_1. \end{aligned}$$

Then,

$$\frac{1}{\sqrt{2}}A(\psi_0 + i\psi_1) = \frac{1}{\sqrt{2}}e^{-2\theta i}(\psi_0 + i\psi_1).$$

Theorem [8.15]

(paperGroverOperator_eigenvectors_of_good_bad_decomposition)

Let $n \in \mathbb{N}$, A be a unitary matrix of dimension 2^n , ψ_0, ψ_1 vectors of dimension 2^n , $f : \mathbb{Z}_{2^n} \rightarrow \{0, 1\}$, and $\theta \in \mathbb{R}$. Assume that $\psi_{0_k} = 0$ for all k such that $f(k) = 1$, while $\psi_{1_k} = 0$ for all k such that $f(k) = 0$. Assume that ψ_0 and ψ_1 are normalized. Assume also that

$$A|0\rangle = \cos(\theta)\psi_0 + \sin(\theta)\psi_1.$$

Then, the following two statements hold.

$$\begin{aligned} \frac{1}{\sqrt{2}}A(\psi_0 - i\psi_1) &= \frac{1}{\sqrt{2}}e^{2\theta i}(\psi_0 - i\psi_1), \\ \frac{1}{\sqrt{2}}A(\psi_0 + i\psi_1) &= \frac{1}{\sqrt{2}}e^{-2\theta i}(\psi_0 + i\psi_1). \end{aligned}$$

Definition [8.9]

(paperQAEInitialState)

Let $m, n \in \mathbb{N}$ and A be a square matrix of dimension 2^n .

$$E_1(m, n, A) = \underbrace{|0 \dots 0\rangle}_m \otimes \underbrace{(A|0 \dots 0\rangle)}_n.$$

Definition [8.10]

(paperQAEAfterQFT)

Let $m, n \in \mathbb{N}$ and A be a square matrix of dimension 2^n .

$$E_2(m, n, A) = (Q(m) \otimes I_n) \times E_1.$$

Definition [8.11]

(paperQAEAfterControlledPowers)

Let $m, n \in \mathbb{N}$, A be a square matrix of dimension 2^n , and $f : \mathbb{Z}_{2^n} \rightarrow \{0, 1\}$.

$$E_3(m, n, A, f) = \Lambda(G_f(A), m) \times E_2.$$

Definition [8.12]

(paperQAEFinalState)

Let $m, n \in \mathbb{N}$, A be a square matrix of dimension 2^n , and $f : \mathbb{Z}_{2^n} \rightarrow \{0, 1\}$.

$$E_4(m, n, A, f) = (IQ(m) \otimes I_n) \times E_3.$$

Definition [8.13]

(paperQAEOutputProbability)

Let $m, n \in \mathbb{N}$, A be a square matrix of dimension 2^n , $f : \mathbb{Z}_{2^n} \rightarrow \{0, 1\}$, and $y \in \mathbb{Z}_{2^m}$.

$$|| \langle y | E_4(m, n, A, f) ||^2 = \sum_{j=0}^{2^n-1} || \langle y, j | E_4(m, n, A, f) ||^2.$$

Definition [8.14]

(paperQAEEstimate)

$$\sin^2 \left(\pi \frac{y}{2^m} \right).$$

Definition [8.15]

(paperPreparedState)

$$A |0\rangle.$$

Definition [8.16]

(paperBadProjection)

$$b(n, A, f) = \sum_{\substack{k=0 \\ f(k)=0}}^{2^n-1} \langle k | A |0\rangle |k\rangle.$$

Definition [8.17]
(paperGoodProjection)

$$g(n, A, f) = \sum_{\substack{k=0 \\ f(k)=1}}^{2^n-1} \langle k | A | 0 \rangle | k \rangle .$$

Definition [8.18]
(paperBadProbability)

$$\sum_{\substack{k=0 \\ f(k)=0}}^{2^n-1} \langle k | A | 0 \rangle \langle 0 | A^\dagger | k \rangle .$$

Definition [8.19]
(paperGoodProbability)

$$\sum_{\substack{k=0 \\ f(k)=1}}^{2^n-1} \langle k | A | 0 \rangle \langle 0 | A^\dagger | k \rangle .$$

Definition [8.20]
(paperBadState)

$$\frac{1}{\cos(\theta)} \sum_{\substack{k=0 \\ f(k)=0}}^{2^n-1} \langle k | A | 0 \rangle | k \rangle .$$

Definition [8.21]
(paperGoodState)

$$\frac{1}{\sin(\theta)} \sum_{\substack{k=0 \\ f(k)=1}}^{2^n-1} \langle k | A | 0 \rangle | k \rangle .$$

Theorem [8.16]**(paperPreparedState_decompose_bad_good)**

Let $n \in \mathbb{N}$, A be a square matrix of dimension 2^n , and $f : \mathbb{Z}_{2^n} \rightarrow \{0, 1\}$.
Then,

$$A|0\rangle = \sum_{\substack{k=0 \\ f(k)=0}}^{2^n-1} \langle k| A |0\rangle |k\rangle + \sum_{\substack{k=0 \\ f(k)=1}}^{2^n-1} \langle k| A |0\rangle |k\rangle.$$

Corollary**(paperPreparedState_decompose_bad_good_states)**

$$A|0\rangle = \cos(\theta) \sum_{\substack{k=0 \\ f(k)=0}}^{2^n-1} \frac{1}{\cos(\theta)} \langle k| A |0\rangle |k\rangle + \sin(\theta) \sum_{\substack{k=0 \\ f(k)=1}}^{2^n-1} \frac{1}{\sin(\theta)} \langle k| A |0\rangle |k\rangle.$$

Theorem [8.17]**(paperBadState_support)**

Let $n \in \mathbb{N}$, A be a square matrix of dimension 2^n , $f : \mathbb{Z}_{2^n} \rightarrow \{0, 1\}$, and $x \in \mathbb{Z}_{2^n}$. Assume that $f(x) = 1$. Then,

$$\left(\frac{1}{\cos(\theta)} \sum_{\substack{k=0 \\ f(k)=0}}^{2^n-1} \langle k| A |0\rangle |k\rangle \right)_x = 0.$$

Theorem [8.18]**(paperGoodState_support)**

Let $n \in \mathbb{N}$, A be a square matrix of dimension 2^n , $f : \mathbb{Z}_{2^n} \rightarrow \{0, 1\}$, and $x \in \mathbb{Z}_{2^n}$. Assume that $f(x) = 0$. Then,

$$\left(\frac{1}{\sin(\theta)} \sum_{\substack{k=0 \\ f(k)=1}}^{2^n-1} \langle k| A |0\rangle |k\rangle \right)_x = 0.$$

Theorem [8.19]**(paperBadProjection_norm)**

Let $n \in \mathbb{N}$, A be a square matrix of dimension 2^n , and $f : \mathbb{Z}_{2^n} \rightarrow \{0, 1\}$. Then,

$$b(n, A, f)^\dagger b(n, A, f) = \sum_{\substack{k=0 \\ f(k)=0}}^{2^n-1} \langle k | A | 0 \rangle \langle 0 | A^\dagger | k \rangle.$$

Theorem [8.20]

(paperGoodProjection_norm)

Let $n \in \mathbb{N}$, A be a square matrix of dimension 2^n , and $f : \mathbb{Z}_{2^n} \rightarrow \{0, 1\}$. Then,

$$g(n, A, f)^\dagger g(n, A, f) = \sum_{\substack{k=0 \\ f(k)=1}}^{2^n-1} \langle k | A | 0 \rangle \langle 0 | A^\dagger | k \rangle.$$

Theorem [8.21]

(paperBadState_isNormalized)

Let $n \in \mathbb{N}$, A be a square matrix of dimension 2^n , $f : \mathbb{Z}_{2^n} \rightarrow \{0, 1\}$, and $\theta \in \mathbb{R}$. Assume that

$$\sum_{\substack{k=0 \\ f(k)=0}}^{2^n-1} \langle k | A | 0 \rangle \langle 0 | A^\dagger | k \rangle = 1 - \sin^2(\theta).$$

Also, assume that $\cos(\theta) \neq 0$. Then, $\frac{1}{\cos(\theta)} b(n, A, f)$ is normalized.

Theorem [8.22]

(paperGoodState_isNormalized)

Let $n \in \mathbb{N}$, A be a square matrix of dimension 2^n , $f : \mathbb{Z}_{2^n} \rightarrow \{0, 1\}$, and $\theta \in \mathbb{R}$. Assume that

$$\sum_{\substack{k=0 \\ f(k)=1}}^{2^n-1} \langle k | A | 0 \rangle \langle 0 | A^\dagger | k \rangle = \sin^2(\theta).$$

Also, assume that $\sin(\theta) \neq 0$. Then, $\frac{1}{\sin(\theta)} g(n, A, f)$ is normalized.

Theorem [8.23]**(paperBadProbability_add_goodProbability)**

Let $n \in \mathbb{N}$, A be a square matrix of dimension 2^n , and $f : \mathbb{Z}_{2^n} \rightarrow \{0, 1\}$.
Then,

$$\sum_{k=0}^{2^n-1} (\langle k | b(n, A, f) | k \rangle + \langle k | g(n, A, f) | k \rangle) = \sum_{k=0}^{2^n-1} \langle k | A | 0 \rangle \langle 0 | A^\dagger | k \rangle.$$

Theorem [8.24]**(paperBadProbability_add_goodProbability_eq_one)**

Let $n \in \mathbb{N}$, A be a unitary matrix of dimension 2^n , and $f : \mathbb{Z}_{2^n} \rightarrow \{0, 1\}$.
Then,

$$\sum_{k=0}^{2^n-1} (\langle k | b(n, A, f) | k \rangle + \langle k | g(n, A, f) | k \rangle) = 1.$$

Corollary**(paperBadProbability_eq_one_sub_goodProbability)**

$$\sum_{k=0}^{2^n-1} \langle k | b(n, A, f) | k \rangle = 1 - \sum_{k=0}^{2^n-1} \langle k | g(n, A, f) | k \rangle.$$

Theorem [8.25]**(paperBadProbability_zero_support)**

Let $n \in \mathbb{N}$, A be a unitary matrix of dimension 2^n , $f : \mathbb{Z}_{2^n} \rightarrow \{0, 1\}$, and $x \in \mathbb{Z}_{2^n}$. Assume that $f(x) = 0$ and that

$$\sum_{\substack{k=0 \\ f(k)=0}}^{2^n-1} \langle k | A | 0 \rangle \langle 0 | A^\dagger | k \rangle = 0.$$

Then,

$$(A | 0 \rangle)_x = 0.$$

Theorem [8.26]

(paperGoodProbability_zero_support)

Let $n \in \mathbb{N}$, A be a unitary matrix of dimension 2^n , $f : \mathbb{Z}_{2^n} \rightarrow \{0, 1\}$, and $x \in \mathbb{Z}_{2^n}$. Assume that $f(x) = 1$ and that

$$\sum_{\substack{k=0 \\ f(k)=1}}^{2^n-1} \langle k | A | 0 \rangle \langle 0 | A^\dagger | k \rangle = 0.$$

Then,

$$(A | 0 \rangle)_x = 0.$$

Theorem [8.27]

(phaseState_zero_eq_uniformState)

Let $m \in \mathbb{N}$. Then,

$$Q(m) |p(m, 0)\rangle = |u(m)\rangle.$$

Theorem [8.28]

(paperQAEAfterQFT_eq_uniformState_tensor)

Let $m, n \in \mathbb{N}$ and A be a square matrix of dimension 2^n . Then,

$$E_2(m, n, A) = |u(m)\rangle \otimes (A \underbrace{|0 \dots 0\rangle}_n).$$

Theorem [8.29]

(paperQAEFinalState_eq_eigenphase_superposition)

Let $m, n \in \mathbb{N}$, A be a square matrix of dimension 2^n , $f : \mathbb{Z}_{2^n} \rightarrow \{0, 1\}$, ψ, ϕ vectors of dimension 2^n , $\theta_\psi, \theta_\phi \in \mathbb{R}$, and $a, b \in \mathbb{C}$. Assume that

$$A | 0 \rangle = a\psi + b\phi.$$

Assume also that the following two propositions hold.

$$\begin{aligned} G_f(A)\psi &= e^{2\pi i\theta_\psi}\psi, \\ G_f(A)\phi &= e^{2\pi i\theta_\phi}\phi. \end{aligned}$$

Then,

$$E_4(m, n, A, f) = aIQ(m)(|p(m, \theta_\psi)\rangle \otimes \psi) + bIQ(m)(|p(m, \theta_\phi)\rangle \otimes \phi).$$

Theorem [8.30]

(paperQAEOutputProbability_eq_partialProb)

Let $n \in \mathbb{N}$, A be a square matrix of dimension 2^n , $f : \mathbb{Z}_{2^n} \rightarrow \{0, 1\}$, and $y \in \mathbb{Z}_{2^m}$. Then,

$$||\langle y | E_4(m, n, A, f) ||^2 = \sum_{j=0}^{2^n-1} ||\langle y, j | E_4(m, n, A, f) ||^2.$$

Theorem [8.31]

(paperQAEOutputProbability_eq_single_eigenphase)

Let $m, n \in \mathbb{N}$, A be a unitary matrix of dimension 2^n , $f : \mathbb{Z}_{2^n} \rightarrow \{0, 1\}$, $y \in \mathbb{Z}_{2^m}$, and $\theta \in \mathbb{R}$. Assume that

$$G_f(A)A|0\rangle = e^{2\pi i\theta}A|0\rangle.$$

Then,

$$||\langle y | E_4(m, n, A, f) ||^2 = ||a(m, \theta, y) ||^2.$$

Theorem [8.32]

(paperQAEOutputProbability_eq_eigenphase_mixture)

Let $m, n \in \mathbb{N}$, A be a square matrix of dimension 2^n , $f : \mathbb{Z}_{2^n} \rightarrow \{0, 1\}$, ψ, ϕ vectors of dimension 2^n , $\theta_\psi, \theta_\phi \in \mathbb{R}$, and $a, b \in \mathbb{C}$. Assume that

$$A|0\rangle = a\psi + b\phi.$$

Assume that the following two propositions hold.

$$\begin{aligned} G_f(A)\psi &= e^{2\pi i\theta_\psi}\psi, \\ G_f(A)\phi &= e^{2\pi i\theta_\phi}\phi. \end{aligned}$$

Also, assume that ψ and ϕ are orthonormal. Let $y \in \mathbb{Z}_{2^m}$. Then,

$$||\langle y | E_4(m, n, A, f) ||^2 = ||a||^2 ||a(m, \theta_\psi, y)||^2 + ||b||^2 ||a(m, \theta_\phi, y)||^2.$$

Theorem [8.33]

(goodBad_decomposition_eq_eigen_superposition)

Let $n \in \mathbb{N}$, ψ_0, ψ_1 be vectors of dimension 2^n , and $\theta \in \mathbb{R}$. Then,

$$\cos(\theta)\psi_0 + \sin(\theta)\psi_1 = \frac{1}{2}(\cos(\theta) + i\sin(\theta))(\psi_0 - i\psi_1) + \frac{1}{2}(\cos(\theta) - i\sin(\theta))(\psi_0 + i\psi_1).$$

Theorem [8.34]

(orthonormal_linear_combination_isNormalized)

Let $n \in \mathbb{N}$, ψ_0, ψ_1 be vectors of dimension 2^n , and $a, b \in \mathbb{C}$. Assume that ψ_0 and ψ_1 are orthonormal. And that $a^\dagger a + b^\dagger b = 1$. Then, $a\psi_0 + b\psi_1$ is normalized.

Theorem [8.35]

(orthonormal_linear_combination_inner)

Let $n \in \mathbb{N}$, ψ_0, ψ_1 be vectors of dimension 2^n , and $a, b, c, d \in \mathbb{C}$. Assume that ψ_0 and ψ_1 are orthonormal. Then,

$$(a\psi_0 + b\psi_1)^\dagger (c\psi_0 + d\psi_1) = a^\dagger c + b^\dagger d.$$

Theorem [8.36]

(eigenMinus_goodBad_isNormalized)

Let $n \in \mathbb{N}$ and ψ_0, ψ_1 be orthonormal vectors of dimension 2^n . Then, the vector $\frac{1}{\sqrt{2}}(\psi_0 - i\psi_1)$ is normalized.

Theorem [8.37]

(eigenPlus_goodBad_isNormalized)

Let $n \in \mathbb{N}$ and ψ_0, ψ_1 be orthonormal vectors of dimension 2^n . Then, the vector $\frac{1}{\sqrt{2}}(\psi_0 + i\psi_1)$ is normalized.

Theorem [8.38]

(eigenMinus_inner_eigenPlus_goodBad)

Let $n \in \mathbb{N}$ and ψ_0, ψ_1 be orthonormal vectors of dimension 2^n . Vectors $\frac{1}{\sqrt{2}}(\psi_0 - i\psi_1)$ and $\frac{1}{\sqrt{2}}(\psi_0 + i\psi_1)$ are orthogonal.

Theorem [8.39]

(paperQAEOutputProbability_eq_good_bad_mixture)

Let $m, n \in \mathbb{N}$, A be a unitary matrix of dimension 2^n , $f : \mathbb{Z}_{2^n} \rightarrow \{0, 1\}$, ψ_0, ψ_1 vectors of dimension 2^n , $\theta \in \mathbb{R}$, and $y \in \mathbb{Z}_{2^m}$. Assume that

$$A |0\rangle = \cos(\theta)\psi_0 + \sin(\theta)\psi_1.$$

Assume also that $\psi_{0k} = 0$ for all k such that $f(k) = 1$, while $\psi_{1k} = 0$ for all k such that $f(k) = 0$. Moreover, assume that ψ_0 and ψ_1 are normalized.

Then,

$$\| \langle y | E_4(m, n, A, f) \| ^2 = \frac{1}{2} \| a(m, \frac{\theta}{\pi}, y) \|^2 + \frac{1}{2} \| a(m, 1 - \frac{\theta}{\pi}, y) \|^2.$$

Definition [8.22]

(qaeGroverPlaneMarginal)

Let $m \in \mathbb{N}$, $\theta \in \mathbb{R}$, $y \in \mathbb{Z}_{2^m}$, and

$$\psi = (IQ(m) \otimes I_2) \times \Lambda(R_y(4\theta), m) \times (|u(m)\rangle \otimes (\cos(\theta) |0\rangle + \sin(\theta) |1\rangle)).$$

$$PM(m, y, \theta) = \sum_{j=0}^1 \| \psi_{j+2y} \|^2.$$

Theorem [8.40]

(qaeGroverPlaneMarginal_eq)

Let $m \in \mathbb{N}$, $\theta \in \mathbb{R}$, $y \in \mathbb{Z}_{2^m}$, and

$$\psi = (IQ(m) \otimes I_2) \times \Lambda(R_y(4\theta), m) \times (|u(m)\rangle \otimes (\cos(\theta) |0\rangle + \sin(\theta) |1\rangle)).$$

Then,

$$\sum_{j=0}^1 ||\psi_{j+2y}||^2 = \frac{1}{2} ||a(m, \frac{\theta}{\pi}, y)||^2 + \frac{1}{2} ||a(m, -\frac{\theta}{\pi}, y)||^2.$$

Theorem [8.41]

(amplitudeFromAngle_pi_sub)

Let $\theta \in \mathbb{R}$. Then,

$$\sin^2(\pi - \theta) = \sin^2(\theta).$$

Theorem [8.42]

(qaeGroverPlaneMarginal_eq_wrapped)

Let $m \in \mathbb{N}$, $\theta \in \mathbb{R}$, $y \in \mathbb{Z}_{2^m}$, and

$$\psi = (IQ(m) \otimes I_2) \times \Lambda(R_y(4\theta), m) \times (|u(m)\rangle \otimes (\cos(\theta) |0\rangle + \sin(\theta) |1\rangle)).$$

Then,

$$\sum_{j=0}^1 ||\psi_{j+2y}||^2 = \frac{1}{2} ||a(m, \frac{\theta}{\pi}, y)||^2 + \frac{1}{2} ||a(m, 1 - \frac{\theta}{\pi}, y)||^2.$$

Theorem [8.43]

(phase_angle_error_of_pos_window)

Let $m, k \in \mathbb{N}$, $\theta \in \mathbb{R}$, and $y \in \mathbb{Z}_{2^m}$. Assume that

$$\left| \frac{\theta}{\pi} - \frac{y}{2^m} \right| \leq \frac{k}{2^m}.$$

Then,

$$\left| \theta - \pi \frac{y}{2^m} \right| \leq \pi \frac{k}{2^m}.$$

Definition [8.23]

(qaeGroverPlaneSuccessfulOutcomesK)

Let $m, k \in \mathbb{N}$, $\theta \in \mathbb{R}$, $a = \sin^2(\theta)$, and $M = 2^m$.

$$S(m, k, \theta) = \left\{ y \in \mathbb{Z}_{2^m} \left| \left| \sin^2 \left(\pi \frac{y}{M} \right) - a \right| \leq 2\pi \frac{k}{M} \sqrt{a(1-a)} + \pi^2 \frac{k^2}{M^2} \right. \right\}$$

Definition [8.24]

(qaeGroverPlaneSuccessProbabilityK)

Let $m, k \in \mathbb{N}$ and $\theta \in \mathbb{R}$.

$$P(m, k, \theta) = \sum_{y \in S(m, k, \theta)} PM(m, y, \theta).$$

Theorem [8.44]

(mem_qaeGroverPlaneSuccessfulOutcomesK_of_estimate_error)

Let $m, k \in \mathbb{N}$, $\theta \in \mathbb{R}$, $y \in \mathbb{Z}_{2^m}$, $a = \sin^2(\theta)$, and $M = 2^m$. Then,

$$y \in S(m, k, \theta) \iff \left| \sin^2 \left(\pi \frac{y}{M} \right) - a \right| \leq 2\pi \frac{k}{M} \sqrt{a(1-a)} + \pi^2 \frac{k^2}{M^2}.$$

Theorem [8.45]

(amplitudeFromAngle_sub_pi)

Let $\theta \in \mathbb{R}$. Then,

$$\sin^2(\theta - \pi) = \sin^2(\theta).$$

Theorem [8.46]

(amplitudeFromAngle_add_pi)

Let $\theta \in \mathbb{R}$. Then,

$$\sin^2(\theta + \pi) = \sin^2(\theta).$$

Theorem [8.47]

(phase_angle_error_of_window_sub_one)

Let $m, k \in \mathbb{N}$, $\alpha \in \mathbb{R}$, and $y \in \mathbb{Z}_{2^m}$. Assume that

$$\left| \frac{\alpha}{\pi} - \frac{y}{2^m} - 1 \right| \leq \frac{k}{2^m}.$$

Then,

$$\left| \pi \frac{y}{2^m} - (\alpha - \pi) \right| \leq \pi \frac{k}{2^m}.$$

Theorem [8.48]

(phase_angle_error_of_window_add_one)

Let $m, k \in \mathbb{N}$, $\alpha \in \mathbb{R}$, and $y \in \mathbb{Z}_{2^m}$. Assume that

$$\left| \frac{\alpha}{\pi} - \frac{y}{2^m} + 1 \right| \leq \frac{k}{2^m}.$$

Then,

$$\left| \pi \frac{y}{2^m} - (\alpha + \pi) \right| \leq \pi \frac{k}{2^m}.$$

Theorem [8.49]

(estAmp_error_of_phase_circular_window)

Let $m, k \in \mathbb{N}$, $\alpha \in \mathbb{R}$, $y \in \mathbb{Z}_{2^m}$, $a = \sin^2(\alpha)$, and $M = 2^m$. Assume that

$$d\left(\frac{\alpha}{\pi}, \frac{y}{2^m}\right) \leq \frac{k}{2^m}.$$

Then,

$$\left| \sin^2\left(\pi \frac{y}{2^m}\right) - a \right| \leq 2\pi \frac{k}{M} \sqrt{a(1-a)} + \pi^2 \frac{k^2}{M^2}.$$

Theorem [8.50]

(qaeGroverPlane_estimate_error_of_wrapped_neg_circular_window)

Let $m, k \in \mathbb{N}$, $\theta \in \mathbb{R}$, $y \in \mathbb{Z}_{2^m}$, $a = \sin^2(\theta)$, and $M = 2^m$. Assume that

$$d\left(1 - \frac{\theta}{\pi}, \frac{y}{2^m}\right) \leq \frac{k}{2^m}.$$

Then,

$$\left| \sin^2 \left(\pi \frac{y}{2^m} \right) - a \right| \leq 2\pi \frac{k}{M} \sqrt{a(1-a)} + \pi^2 \frac{k^2}{M^2}.$$

Theorem [8.51]

(qaeGroverPlane_pos_circular_window_subset_success)

Let $m, k \in \mathbb{N}$ and $\theta \in \mathbb{R}$. Then,

$$O_{m,k,\frac{\theta}{\pi}} \subseteq S_{m,k,\theta}.$$

Corollary

(qaeGroverPlane_wrapped_neg_circular_window_subset_success)

The following holds.

$$O_{m,k,1-\frac{\theta}{\pi}} \subseteq S_{m,k,\theta}.$$

Theorem [8.52]

(qaeGroverPlaneSuccessProbabilityK_lower_bound_of_qpe_circular_windows)

Let $m, k \in \mathbb{N}$ and $\theta \in \mathbb{R}$. Assume that the following two propositions hold.

$$\begin{aligned} \begin{cases} \frac{8}{\pi^2}, & k = 1 \\ 1 - \frac{1}{2(k-1)}, & \text{otherwise} \end{cases} &\leq \Pr\{y \in O_{m,k,\frac{\theta}{\pi}} | Y\}, \\ \begin{cases} \frac{8}{\pi^2}, & k = 1 \\ 1 - \frac{1}{2(k-1)}, & \text{otherwise} \end{cases} &\leq \Pr\{y \in O_{m,k,1-\frac{\theta}{\pi}} | Y\}. \end{aligned}$$

Then,

$$\begin{cases} \frac{8}{\pi^2}, & k = 1 \\ 1 - \frac{1}{2(k-1)}, & \text{otherwise} \end{cases} \leq P(m, k, \theta).$$

Theorem [8.53]

(qaeGroverPlaneSuccessProbabilityK_lower_bound)

Let $m, k \in \mathbb{N}$ and $\theta \in [0, \pi]$. Assume that $k > 0$. Then,

$$\begin{cases} \frac{8}{\pi^2}, & k = 1 \\ 1 - \frac{1}{2(k-1)}, & \text{otherwise} \end{cases} \leq P(m, k, \theta).$$

Theorem [8.54]

(theorem12ErrorBound_eq_expanded_QPE)

Just a simple unfolding of the definition of theorem12ErrorBound.

Theorem [8.55]

(paperQAEsSuccessProbabilityK_lower_bound)

Let $m, n, k \in \mathbb{N}$, A be a unitary matrix of dimension 2^n , $f : \mathbb{Z}_{2^n} \rightarrow \{0, 1\}$, ψ_0, ψ_1 vectors of dimension 2^n , and $\theta \in [0, \pi]$. Assume that $k > 0$, ψ_0 and ψ_1 are normalized, that $\psi_{0k} = 0$ for all k such that $f(k) = 1$, and $\psi_{1k} = 0$ for all k such that $f(k) = 0$. Assume also that

$$A|0\rangle = \cos(\theta)\psi_0 + \sin(\theta)\psi_1.$$

Then,

$$\begin{cases} \frac{8}{\pi^2}, & k = 1 \\ 1 - \frac{1}{2(k-1)}, & \text{otherwise} \end{cases} \leq \sum_{y \in S(m, k, \theta)} ||\langle y | E_4(m, n, A, f) ||^2.$$

Corollary

(paperQAEsSuccessProbabilityK_lower_bound_expanded)

The same as the theorem above, except that $S(m, k, \theta)$ is unfolded.

Theorem [8.56]

(qpeApproxOutcomeProbability_exact_grid_eq_one)

Let $m \in \mathbb{N}$ and $y \in \mathbb{Z}_{2^m}$. Then,

$$||a(m, \frac{y}{2^m}, y)||^2 = 1.$$

Theorem [8.57]

(paperQAEHalfGridRatio)

Let $m \in \mathbb{N}$ such that $m > 0$. Then,

$$\frac{2^{m-1}}{2^m} = \frac{1}{2}.$$

Theorem [8.58]

(paperGroverOperator_mul_prepared_of_goodProbability_zero)

Let $n \in \mathbb{N}$, A be a unitary matrix of dimension 2^n , and $f : \mathbb{Z}_{2^n} \rightarrow \{0, 1\}$. Assume that

$$\sum_{\substack{k=0 \\ f(k)=1}}^{2^n-1} \langle k | A | 0 \rangle \langle 0 | A^\dagger | k \rangle = 0.$$

Then,

$$G_f(A) \times A | 0 \rangle = A | 0 \rangle.$$

Theorem [8.59]

(paperGroverOperator_mul_prepared_of_badProbability_zero)

Let $n \in \mathbb{N}$, A be a unitary matrix of dimension 2^n , and $f : \mathbb{Z}_{2^n} \rightarrow \{0, 1\}$. Assume that

$$\sum_{\substack{k=0 \\ f(k)=0}}^{2^n-1} \langle k | A | 0 \rangle \langle 0 | A^\dagger | k \rangle = 0.$$

Then,

$$G_f(A) \times A | 0 \rangle = e^{\pi i} A | 0 \rangle.$$

Theorem [8.60]

(paperQAEEndpointZero_of_goodProbability_zero)

Let $n \in \mathbb{N}$, A be a unitary matrix of dimension 2^n , and $f : \mathbb{Z}_{2^n} \rightarrow \{0, 1\}$. Assume that

$$\sum_{\substack{k=0 \\ f(k)=1}}^{2^n-1} \langle k | A | 0 \rangle \langle 0 | A^\dagger | k \rangle = 0.$$

Then, there exists y such that

$$|| \langle y | E_4(m, n, A, f) ||^2 = 1.$$

Moreover, for such y ,

$$\sin^2 \left(\pi \frac{y}{2^m} \right) = 0.$$

Theorem [8.61]

(paperQAEEndpointOne_of_badProbability_zero)

Let $m, n \in \mathbb{N}$, A be a unitary matrix of dimension 2^n , and $f : \mathbb{Z}_{2^n} \rightarrow \{0, 1\}$.

Assume that $m > 0$ and that

$$\sum_{\substack{k=0 \\ f(k)=0}}^{2^n-1} \langle k | A | 0 \rangle \langle 0 | A^\dagger | k \rangle = 0.$$

Then, there exists y such that

$$|| \langle y | E_4(m, n, A, f) ||^2 = 1.$$

Moreover, for such y ,

$$\sin^2 \left(\pi \frac{y}{2^m} \right) = 1.$$

Theorem [8.62]

(paperQAEOutputProbability_nonneg)

Let $m, n \in \mathbb{N}$, A be a square matrix of dimension 2^n , $f : \mathbb{Z}_{2^n} \rightarrow \{0, 1\}$, and $y \in \mathbb{Z}_{2^m}$. Then,

$$|| \langle y | E_4(m, n, A, f) ||^2 \geq 0.$$

Theorem [8.63]

(paperQAESuccessSum_ge_one_of_point)

Let $m, n, k \in \mathbb{N}$, A be a square matrix of dimension 2^n , $f : \mathbb{Z}_{2^n} \rightarrow \{0, 1\}$, $a \in \mathbb{R}$, $M = 2^m$, and $y \in \mathbb{Z}_{2^m}$. Assume that

$$|| \langle y | E_4(m, n, A, f) ||^2 = 1.$$

Moreover, assume that

$$|\sin^2\left(\pi \frac{y}{2^m}\right) - a| \leq 2\pi \frac{k}{M} \sqrt{a(1-a)} + \pi^2 \frac{k^2}{M^2}.$$

Then,

$$1 \leq \sum_{y \in S(m,k,\theta)} ||\langle y| E_4(m, n, A, f) ||^2.$$

Theorem [8.64]

(theorem12SuccessProbability_le_one)

Let $k \in \mathbb{N}$ such that $k > 0$, then

$$\begin{cases} \frac{8}{\pi^2}, & k = 1 \\ 1 - \frac{1}{2(k-1)}, & \text{otherwise} \end{cases} \leq 1.$$

Theorem [8.65]

(PaperTheorem12)

Let $m, n, k \in \mathbb{N}$, A be a unitary matrix of dimension 2^n , $f : \mathbb{Z}_{2^n} \rightarrow \{0, 1\}$, $\theta \in [0, \pi]$, $a = \sin^2(\theta)$, $m > 0$, and $k > 0$. Assume that

$$\sum_{\substack{k=0 \\ f(k)=1}}^{2^n-1} \langle k| A |0\rangle \langle 0| A^\dagger |k\rangle = a.$$

Then, the following four propositions hold.

1. If $k = 1$, then

$$\frac{8}{\pi^2} \leq \sum_{y \in S(m,k,\theta)} ||\langle y| E_4(m, n, A, f) ||^2.$$

2. If $k > 1$, then

$$1 - \frac{1}{2(k-1)} \leq \sum_{y \in S(m,k,\theta)} ||\langle y| E_4(m, n, A, f) ||^2.$$

3. If $a = 0$, then there exists $y \in \mathbb{Z}_{2^m}$ such that

$$||\langle y | E_4(m, n, A, f) \rangle||^2 = 1.$$

Moreover, for such y , $\sin^2(\pi \frac{y}{2^m}) = 0$.

4. If $a = 1$, then there exists $y \in \mathbb{Z}_{2^m}$ such that

$$||\langle y | E_4(m, n, A, f) \rangle||^2 = 1.$$

Moreover, for such y , $\sin^2(\pi \frac{y}{2^m}) = 1$.